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COMP1860 Building our Digital World: Computer Systems & Architecture

S1.W4.L2: Karnaugh Maps

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Learning Outcomes

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1. Apply minimisation of Boolean expressions to simplify logic circuits

Text Book



- This lecture covers material similar to chapter 12 of **Stallings. *Computer Organisation & Architecture***
  - It covers Karnaugh maps (section 12.3, pages 421 to 425)

Minimising Circuits



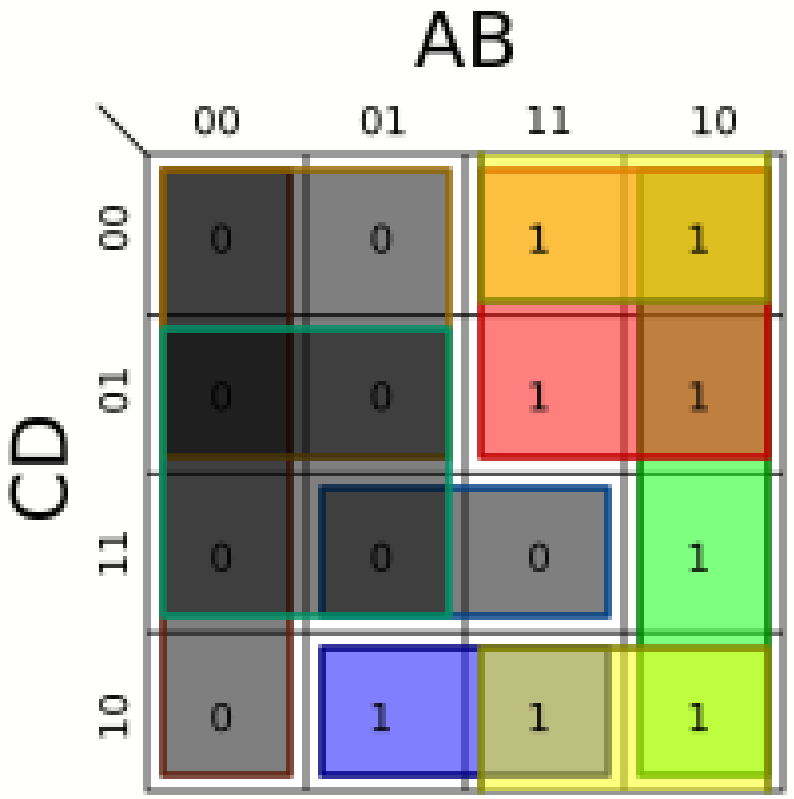
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- To reduce the number of components
- Same as minimising logical function

Karnaugh Maps(K-Map)

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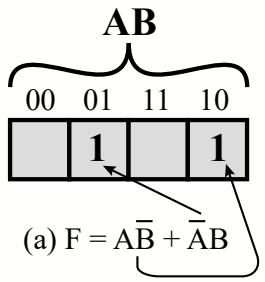
- Another way to simplify Boolean expressions
- Map is similar to truth table
- For n variables, map has 2^n squares
- Note the labelling: 00, 01, 11, 10
  - If you label differently, your map will not produce the correct minimisation!

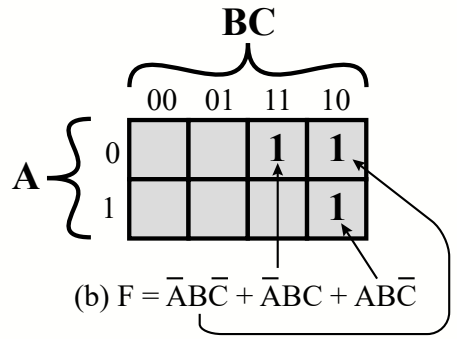


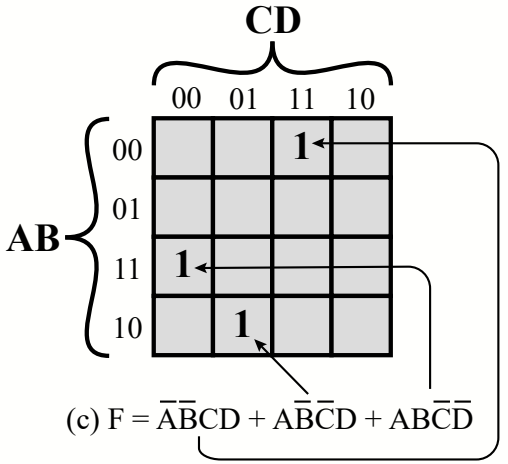
Creating a Map

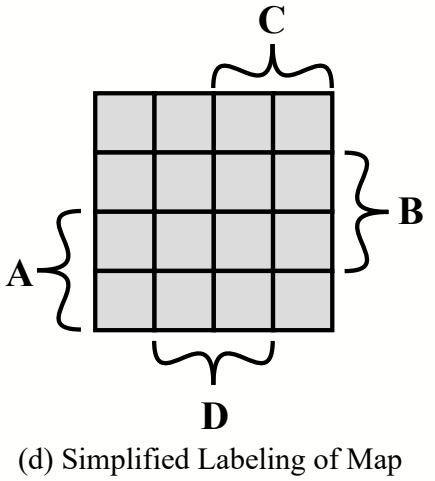
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- 2 variables (A, B)
  - Single row of four entries
  - Four possible values of A and B
- 3 variables (A, B, C)
  - Rows are the two values of variable A
  - Columns are the four possible values of B & C
- 4 variables (A, B, C, D)
  - Rows are the four possible values of A & B
  - Columns are the four possible values of C & D
- Note that we use the sequence 00, 01 11, 10 for 2 values







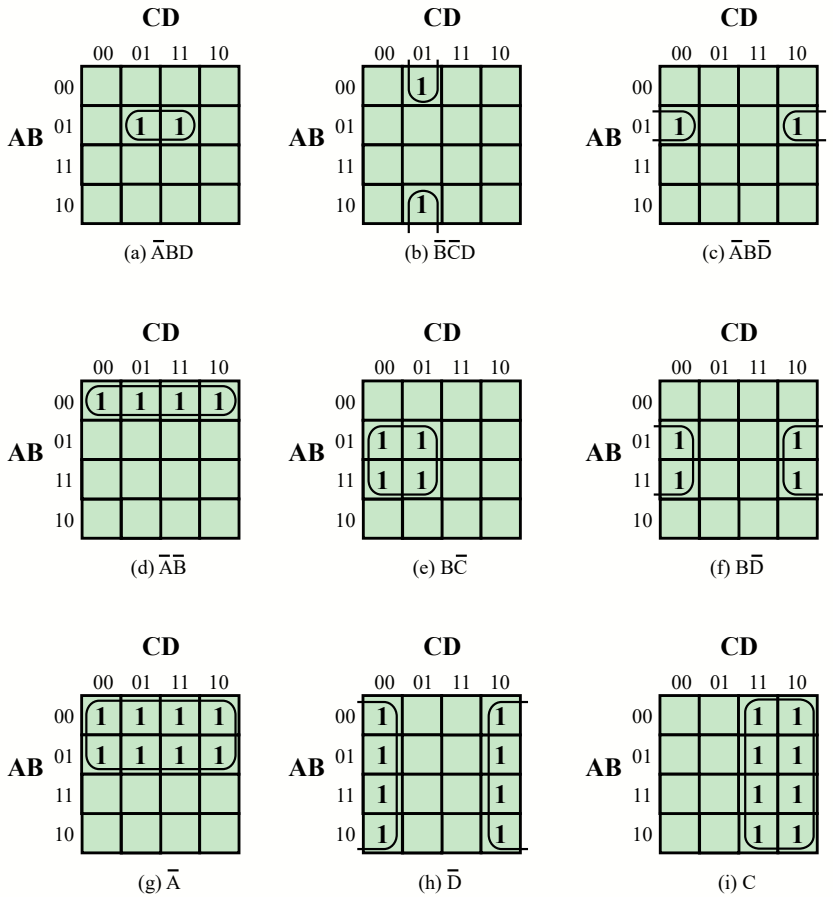


(d) Simplified Labeling of Map

Minimising Using Karnaugh Maps (DNF)

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- Look for groups of adjacent 1's in map
  - Single 1s
  - Lines,Squares or rectangles of 1's
    (水平垂直单行、正方形或矩形)
  - Groups of size 2^n only: 1, 2, 4, 8, …
    (组大小只能是2的指数)
  - Top/bottom, left/right sides wrap around (see figures b, c, f, h) when creating groups
- Try to make groups as large as possible
  (搜索尽可能大的组)



Minimising Using Karnaugh Maps (DNF 1+1)



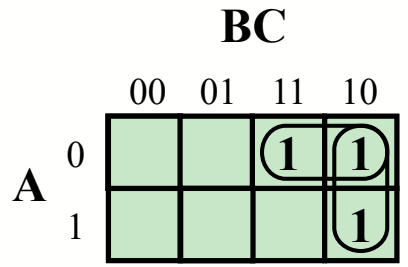
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- Determine which variables have no impact on group
  - The value in map stays the same whether the variable is 0 or 1
  - Ignore that variable
- For remaining inputs create an AND term which evaluates to 1
  - If variable value for group is 1, use as is. E.g., $A = 1$, then A is variable in term
  - If variable value for group is 0, negate. E.g., $A = 0$, then $\neg A$ is variable in term
  - Term is an AND, e.g., $A = 1, B = 0$ would give the term $A \wedge \neg B$
- Terms are joined together by OR.
  - E.g., term from group 1 is $A \wedge \neg B$, term from group 2 is $B \wedge C$, term from group 3 is $\neg A \wedge C$.
  - Hence, minimised expression is $(A \wedge \neg B) \vee (B \wedge C) \vee (\neg A \wedge C)$

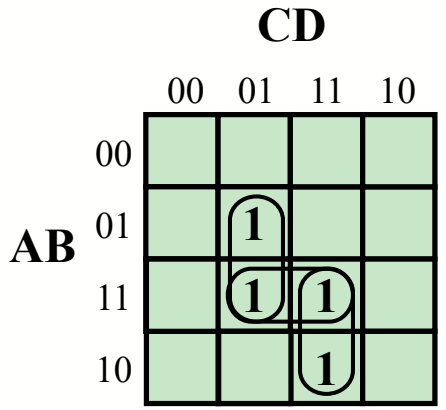
Overlapping Groups(覆盖组)

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- If you have overlapping groups
- Find the minimum number of largest groups to cover all 1's (DNF) or 0's (CNF)



(a) $F = \bar{A}B + B\bar{C}$

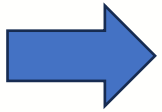


(b) $F = B\bar{C}D + ACD$

Example 1: Create Map

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A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- Arrange truth table as rectangular map
- Note: Different labels on map. Be careful you transfer values correctly!

Example 1: DNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- Circle adjacent or single 1's （圈出相邻或单个1）
  - Try to create largest groups of size 2^n
  - we can create three groups of 2.
- Blue and orange groups cover all 1's（蓝色和橙色组覆盖所有1）
  - The green group overlaps blue and orange（绿色组与蓝色和橙色重叠）
  - We would normally keep this group
  - However, the 1's in the green group are in the blue and orange groups.

Example 1: DNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- For the blue group:
- A=0, B=0 or 1, and C=1 (two values of BC, 01 and 11)
- Values in group are 1, regardless whether B=0 or 1
  - Therefore, variable B has no effect on the value of the expression
  - Can ignore B
  - Leaves A and C as part of minimised term.

- The minimised term is of form $v_1 \wedge v_2$
  - For A=0 and C=1, if $v_1 \wedge v_2 = 1$, we would need to set $v_1 = \neg A$ and $v_2 = C$. i.e., $\neg A \wedge C$.

Why can we ignore variables in group?

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- Remember previous method to determine DNF(1+1) expression
  - Pick out 1s
  - If variable is 1, the use as is, if 0 then invert
- So, for the blue group, first 1 gives $\neg A \wedge \neg B \wedge C$
- Second 1 gives $\neg A \wedge B \wedge C$
- Combining gives $(\neg A \wedge \neg B \wedge C) \vee (\neg A \wedge B \wedge C)$
- Simplifying gives $(\neg A \wedge C) \wedge (B \vee \neg B) = (\neg A \wedge C) \wedge 1 = (\neg A \wedge C)$
- So, B is not part of minimal term
- Similar for CNF

Example 1: DNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- For the orange group
  - Regardless of whether C is 0 or 1, the map entry is 1
    - Can ignore C
  - A=1 and B=1, to make this 1 we would need $A \wedge B$

Example 1: DNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- Hence our minimised DNF form is:

$$(\neg A \wedge C) \vee (A \wedge B)$$

Example 2: DNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- Circle adjacent or single 1's
  - Try to create largest groups of size 2^n
  - While we could circle each of the 1's, we can create two groups of 2.

Example 2: DNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- For the blue group:
- A=0, B=1, and C= 0 or 1 (two values of BC, 11 and 10)
- Values in group are 1, regardless whether C=0 or 1
  - Therefore, variable C has no effect on the value of the expression
  - Can ignore C

- The minimised term is $\neg A \wedge B$.

Example 2: DNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- For the green group
  - Regardless of whether A is 0 or 1, the map entry is 1
    - Can ignore A
  - B=1 and C=0, to make this 1 we would need $B \wedge \neg C$

Example 2: DNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- Hence our minimised DNF form is:

$$(\neg A \wedge B) \vee (B \wedge \neg C)$$

Why can we use the overlapping one twice in minimisation?(为什么可使用两次重叠来最小化？)

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- Remember previous method to determine DNF expression
  - Pick out 1s
  - If variable is 1, the use as is, if 0 then invert
- First 1 gives $\neg A \wedge B \wedge C$
- Second 1 gives $\neg A \wedge B \wedge \neg C$
- Third 1 give $A \wedge B \wedge \neg C$
- Combining gives $(\neg A \wedge B \wedge C) \vee (\neg A \wedge B \wedge \neg C) \vee (A \wedge B \wedge \neg C)$
- Simplifying gives $(\neg A \wedge B) \wedge (C \vee \neg C) \vee (A \wedge B \wedge \neg C)$
- $= (\neg A \wedge B) \wedge 1 \vee (A \wedge B \wedge \neg C) = (\neg A \wedge B) \vee (A \wedge B \wedge \neg C)$!!

However

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- Remember the algebraic law $A \vee A = A$
- So, $(\neg A \wedge B \wedge C) \vee (\neg A \wedge B \wedge \neg C) \vee (A \wedge B \wedge \neg C)$
- $= (\neg A \wedge B \wedge C) \vee (\neg A \wedge B \wedge \neg C) \vee (\neg A \wedge B \wedge \neg C) \vee (A \wedge B \wedge \neg C)$
- Simplifying gives $((\neg A \wedge B) \wedge (C \vee \neg C)) \vee ((B \wedge \neg C) \wedge (A \vee \neg A))$
- $= (\neg A \wedge B) \vee (B \wedge \neg C)$
- Gives the same result as map
- More complicated derivation than using map!
  - Also, map tells you exactly which terms can be repeated

Example 3: DNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- Enclose groups of 1's
- Note largest groups of adjacent 1's are group of 8 and group of 2.

Example 3: DNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- Blue group
  - A = 1, B = 1, C = 0 or 1, D = 0
  - Independent of C
  - A = 1, B = 1, D =0, so term is $A \wedge B \wedge \neg D$

Example 3: DNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- Orange group
  - A = 0 or 1, B = 0 or 1, C = 1, D = 0 or 1.
  - Value is 1 independent of A, B and D input.
  - C=1, so term is C

Example 3: DNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- DNF:

$$(A \wedge B \wedge \neg D) \vee C$$

Exercise 1

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- Minimize the Boolean expressions represented by the following Karnaugh Maps in DNF form:

1.

A \ BC	00	01	11	10
0	1	0	0	1
1	0	1	1	1

2.

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	1	1
10	0	0	1	1

Minimising Using Karnaugh Maps (CNF 0*0)



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- Look for adjacent or single 0's in map
  - Same process as for grouping 1's in DNF
- Determine which variables have no impact on group
  - The value in the map stays the same whether the variable is 0 or 1
  - Ignore that variable
- For remaining inputs provide an OR term which evaluates to 0
  - If variable value for group is 0, use as is. E.g., $A = 0$, then A is variable in term
  - If variable value for group is 1, negate. E.g., $A = 1$, then $\neg A$ is variable in term
  - Term is an OR, e.g., $A = 0$, $B = 1$ would give the term $A \vee \neg B$.
- Terms are joined together by AND.
  - E.g., term from group 1 is $A \vee \neg B$, term from group 2 is $B \vee C$, term from group 3 is $\neg A \vee C$.
  - Hence, minimised expression is $(A \vee \neg B) \wedge (B \vee C) \wedge (\neg A \vee C)$

Example 1: CNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- Circle adjacent or single 0's
  - Might seem like top right 0 is single.
  - However, it wraps to the top left 0 to create a group of 2.
- Blue and green groups covers all zeroes.

Example 1: CNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- For blue group:
  - B has no effect on map entry values in group, 0 if B=0 or B=1
- For CNF, each minimised term is of form v_1 V v_2
- For the values of A & C in the blue group:
  - To make v_1 V $v_2 = 0$, if $A = 0$ and $C = 0$, then as 0 V 0 = 0, we set $v_1 = A$ and $v_2 = C$.
  - Hence, minimised term is $A \vee C$. （注意：CNF 0*0：选0行；输入是0为正，1为反）

Example 1: CNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- For green group:
  - $A = 1, B = 0, C = 0$ or $C = 1$
  - Map entry has values of 0 regardless whether $C = 0$ or $C = 1$. Ignore C.
  - $A = 1$ and $B = 0$. Therefore, to make $v_1 \vee v_2 = 0$ we require $v_1 = \neg A$ and $v_2 = B$.
  - So, minimised term is $\neg A \vee B$.

Example 1: CNF form

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A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

- Hence our minimised CNF form is:

$$(A \vee C) \wedge (\neg A \vee B)$$

Example 2: CNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- Circle adjacent or single 0's
  - Might seem like furthest right 0 is single.
  - However, it can be made part of a group of 2 by overlapping.

Example 2: CNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- For blue group:
  - A has no effect on map entry, values in group, 0 if A=0 or A=1. Ignore A
  - C has no effect on map entry, values in group, 0 if C=0 or C=1. Ignore C.
- For blue group, B=0. Therefore, to make $v_1 = 0$ we require $v_1 = B$. Hence, term is B.

Example 2: CNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- For orange group:
  - $A = 1, B = 0$ or $B = 1$, and $C = 1$
  - Map entry has values of 0 regardless whether $B = 0$ or $B = 1$. Ignore B.
  - $A = 1$ and C$= 1$. Therefore, to make $v_1 \vee v_2 = 0$ we require $v_1 = \neg A$ and $v_2 = \neg C$.
  - So, minimised term is $\neg A \vee \neg C$.

Example 2: CNF form

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A \ BC	00	01	11	10
0	0	0	1	1
1	0	0	0	1

- Hence our minimised CNF form is:

$$B \wedge (\neg A \vee \neg C)$$

Example 3: CNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- Enclose groups of 0’s

Example 3: CNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- Blue group
  - Stays zero independent of B or D
  - For group, A=0, C=0. Hence, term is $A \vee C$

Example 3: CNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- Orange group
  - Stays zero independent of A or B
  - For group C=0, D=1. Hence, term is $C \vee \neg D$

Example 3: CNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- Green group
  - Stays zero independent of A & D
  - For group, B=0, C=0. Hence, term is $B \vee C$

Example 3: CNF

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AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	1
10	0	0	1	1

- CNF:

$$(A \vee C) \wedge (C \vee \neg D) \wedge (B \vee C)$$

Exercise 2

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- Minimize the Boolean expressions represented by the following Karnaugh Maps in CNF form:

1.

A \ BC	00	01	11	10
0	1	0	0	1
1	0	1	1	1

2.

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	1	1
10	0	0	1	1